

Quoting is Colour-Swap: a model of the rho calculus in the knotted universe

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Abstract

The knotted universe of *A Knotted Universe: a new notion of reflective set theories* closes with a conjecture: that the reflective higher-order (rho) calculus has a natural set-theoretic model there, with the quote $@$ read as sealing a process into an atom of the dual colour and the dereference $*$ as unsealing it. We make the conjecture good, and in a form that puts the colour barrier itself to work. We *stratify* the universe into four sorts — red sets and red atoms, black sets and black atoms — linked by the two knots and by a single polymorphic involution, the colour-swap s . The calculus’s reflection operators are then *derived*: $@ = \kappa \circ s$ and $* = s \circ \kappa^{-1}$, so that $*@P \equiv P$ becomes $s \circ s = \text{id}$ — the reflection law is the involutivity of the swap, not an axiom about the model. We give the dynamics following Milner’s guidance for the polyadic π -calculus: a process denotes the set of its *commitments*, input committing to an abstraction $(y)P$ and output to a concretion $[Q]$, with communication the application $(y)P \cdot [M] = P\{M/y\}$ — and it is exactly here, in the reaction rule, that the swap fires, sealing the sent process into the received name. The denotation lands in the final coalgebra of a behaviour functor over the knotted nominal universe, where the colour-blind non-well-foundedness supplies recursion for free and finality makes bisimilar processes equal. Following the GSLT discipline we label transitions not by primitive actions but by *contexts* — the minimal environments enabling a reaction, after Leifer and Milner — and express bisimulation over these context labels; the labels are the program/environment cuts, simultaneously the indices of the OSLF-generated modal logic. We prove a *full abstraction* theorem, $\llbracket P \rrbracket = \llbracket Q \rrbracket$ iff $P \sim Q$, by identifying context bisimilarity with the kernel of the unique morphism to the final coalgebra; because the labels are contexts, the resulting equivalence is a congruence by construction, so full abstraction holds for a congruence with no closure step. The risk ledger of the source paper transfers: finite processes cost no infinite risk; recursion — which in rho enters through reflection, not a primitive replication — costs one ω , black infinity, and not one primitive name. Aczel’s CCS model is recovered as the equivariant, colour-respecting fragment.

1 Introduction

Aczel modelled Milner’s CCS inside non-well-founded set theory [1]: a process P is identified with the set of its labelled transitions $\{(a, P') : P \xrightarrow{a} P'\}$, a looping process is a non-well-founded set, bisimilar processes are *equal*, and the anti-foundation axiom’s solution lemma individuates processes as the unique solutions of their behavioural equations. The construction is canonical, but it buys two things by fiat: an external, unstructured supply of action names a , and the non-well-foundedness itself.

The reflective higher-order calculus — the rho calculus [3] — differs from CCS and the π -calculus precisely where Aczel spends his first postulate. Rho has *no* primitive names. A name is a quoted process: the quote $@$ turns a process into a name, the dereference $*$ turns a name back into the

process it quotes, and the two compose to the identity, $*@P \equiv P$. There is correspondingly no restriction operator ν and no scope extrusion: fresh names are *computed*, by quoting.

The knotted universe of [2] was built from the empty set alone, as two mutually recursive copies of set theory — a red copy $\text{Set}[X]$ and a black copy $\text{Set}[\bar{X}]$ — tied by $X = \text{Set}[\bar{X}]$, $\bar{X} = \text{Set}[X]$, so that each colour’s *atoms* are the other colour’s *sets*. Section 9.3 of [2] conjectures this universe is the natural home for an Aczel-style model of rho, with $@$ sealing a process into an atom of the opposite colour and $*$ unsealing it. We carry the conjecture out. Three design decisions shape the development.

Quoting is the colour-swap, read through a stratification (Section 3). Rather than collapse “red atom” and “black set” into one sort, as the bare knot does, we keep four sorts and make the colour-moving maps first class. A single polymorphic involution $\mathbf{s}$, the colour-swap, exchanges red and black *while preserving the set/atom distinction*; the two knots $\kappa, \bar{\kappa}$ supply the set/atom coercions. The calculus’s $@$ and $*$ are then *composites* of these — $@ = \kappa \circ \mathbf{s}$, $* = \mathbf{s} \circ \kappa^{-1}$ — and the reflection law is $\mathbf{s} \circ \mathbf{s} = \text{id}$. This is at once more honest (the colour crossing is explicit, not hidden in a definitional identity) and exactly the graded, colour-tracking discipline [2], §6.5 asks for: the four sorts are the grading, and $\mathbf{s}$ is the reflective move.

The dynamics follow Milner (Section 5). After Milner’s polyadic π tutorial [4], a process denotes the set of its immediate *commitments*; input commits to an *abstraction* $(y)P$ and output to a *concretion* $[Q]$, and communication is the application $(y)P \cdot [M] = P\{ @M/y \}$. The reaction rule is where the swap of Section 3 does its dynamical work: it seals the sent process M into the name $@M$ the receiver binds. Parallel composition is not bare union — that would wrongly identify $a!(0) | a!(0)$ with $a!(0)$ — but the corecursive expansion $\parallel$ that keeps each component’s surviving context in the residuals.

Transitions are labelled by contexts (Sections 2.3, 6). To stay consistent with the GSLT programme we do not label transitions by primitive actions but by *contexts*: $P \xrightarrow{F} P'$ means F is a minimal environment — an idem-pushout, after Leifer and Milner [5, 6], adapted to GSLTs [7] — in which P completes a reaction to P' . The label is the program/environment cut at which P meets its world, and it is at once the Firing label of the transition system, the Witnessing index of an OSLF-generated Hennessy–Milner modality $\langle F \rangle \varphi$, and the Placing of the interaction in P ’s term structure — three views of one ∂T . Expressing bisimulation over context labels has a structural dividend: the equivalence is a congruence by construction, so the full-abstraction theorem holds for a congruence with none of the closure-under-substitution care an action-labelled account requires.

Recursion is free (Sections 4, 6). The denotation lands in the final coalgebra of a behaviour functor over the knotted nominal universe; the colour-blind non-well-foundedness of [2], §7 supplies the recursion (which in rho is reached through reflection, with no primitive replication operator), and finality makes bisimilar processes equal. We prove full abstraction, $\llbracket P \rrbracket = \llbracket Q \rrbracket \iff P \sim Q$, identifying context bisimilarity with the kernel of the unique morphism to the final coalgebra, and read it back through OSLF adequacy.

We work throughout with the modern presentation: $@P$ for quoting, $*x$ for dereference, $\text{for}(y \leftarrow x) P$ for input.

2 The rho calculus

2.1 Syntax

The two-sorted grammar of processes P and names x :

$$P, Q ::= \mathbf{0} \mid P \mid Q \mid \text{for}(y \leftarrow x) P \mid x!(Q) \mid *x, \quad x, y ::= @P.$$

$\mathbf{0}$ is the stopped process; $P \mid Q$ parallel composition; $\text{for}(y \leftarrow x) P$ the input receiving on channel x , binding the received name to y in P ; $x!(Q)$ the asynchronous output sending Q on x ; and $*x$ the dereference. A name is solely a quoted process: no primitive names, and no restriction ν . In $\text{for}(y \leftarrow x) P$ the name y is bound in P . Capture-avoiding substitution $P\{x/y\}$ of a name for a free name reaches inside quotes.

There is no primitive replication or recursion. Both are nonetheless available, by reflection: a recursive process is written by quoting its own continuation and dereferencing it, e.g. a server $D \equiv \text{for}(y \leftarrow x) (*y \mid x!(y)) \mid x!(@(\dots))$ reinstalls itself after each interaction. Infinite behaviour from a finite term is therefore present already in the core calculus — which is why the model needs the non-well-founded reading even with no $!$ in the grammar.

2.2 Structural congruence and name equivalence

$\equiv$ is the least congruence closed under α -equivalence and

$$(P \mid Q) \mid R \equiv P \mid (Q \mid R), \quad P \mid Q \equiv Q \mid P, \quad P \mid \mathbf{0} \equiv P, \quad *@P \equiv P.$$

Name equivalence is induced by dereference: $@P \equiv_{\mathcal{N}} @Q :\iff *@P \equiv *@Q \iff P \equiv Q$.

2.3 Reaction and the context-labelled transition system

The base interaction is communication, with the structural propagation rules:

$$\frac{x \equiv_{\mathcal{N}} z}{\text{for}(y \leftarrow x) P \mid z!(Q) \longrightarrow P\{@Q/y\}} \text{COMM} \qquad \frac{P \longrightarrow P'}{P \mid R \longrightarrow P' \mid R} \text{PAR}$$

$$\frac{P \equiv P' \quad P' \longrightarrow Q' \quad Q' \equiv Q}{P \longrightarrow Q} \text{STRUCT}$$

Rho is an interactive GSLT whose single base constructor is parallel composition: $K = \mid$ is the head of the left-hand side of the one hypothesis-free rule, COMM, and the cut it induces is *relative* — each parallel component is, from the other's vantage, the environment.

To define bisimulation we follow the GSLT discipline and label transitions by *contexts* rather than by primitive actions, after the minimal-context (relative-pushout) construction of Leifer and Milner [5, 6], as adapted to GSLTs [7]. A *context* F is a process with one hole $[\cdot]$; $F[P]$ is the result of plugging P . Communication never fires under a prefix, so the reactive contexts are, modulo $\equiv$, the parallel contexts $[\cdot] \mid R$: the cut always lies at a parallel composition, exactly the program/environment cut.

Definition 1 (Context-labelled transitions). $P \xrightarrow{F} P'$ holds when F is a context with $F[P] \longrightarrow P'$ by a single COMM, and F is *minimal* — an idem-pushout, supplying just the missing half of the redex and no more. Concretely, modulo $\equiv$:

- (*offer input*) if $P \equiv \text{for}(y \leftarrow x) P_0 \mid P_1$ then $P \xrightarrow{[\cdot] \mid x!(Q)} P_0\{\text{@}Q/y\} \mid P_1$, one transition per message Q ;
- (*offer output*) if $P \equiv x!(Q) \mid P_1$ then $P \xrightarrow{[\cdot] \mid \text{for}(y \leftarrow x) R} P_1 \mid R\{\text{@}Q/y\}$, one per input context $\text{for}(y \leftarrow x) R$;
- (*internal redex*) if $P \equiv \text{for}(y \leftarrow x) P_0 \mid x!(Q) \mid P_1$ then $P \xrightarrow{[\cdot]} P_0\{\text{@}Q/y\} \mid P_1$.

The label records what environment P needs to interact; the identity context $[\cdot]$ labels the reactions P performs alone (the silent moves). The seal $\text{@} = \kappa \circ \mathbf{s}$ acts in each substitution: communication delivers the *name* $\text{@}Q$ of the sent process, never the process itself.

Remark 2 (The label set is ∂T , in its three views). The labels are the Fire-sub-bundle ∂T_K of the splitting space $\partial T \times T$: the splittings (F, t) whose subterm t completes a COMM redex. This one set is read three ways [7]: as the label set of the transition system (*Firing*); as the index set of the OSLF-generated Hennessy–Milner modalities $\langle F \rangle \varphi$ — “some F -labelled transition reaches a state satisfying φ ” (*Witnessing*); and as the program/environment cuts, i.e. the locations at which P meets its environment (*Placing*). We will use the first to define bisimulation and the second to characterise it.

Definition 3 (Context bisimulation). A symmetric relation $\mathcal{R}$ on processes is a *context bisimulation* if $P \mathcal{R} Q$ implies: whenever $P \xrightarrow{F} P'$ there is $Q \xrightarrow{F} Q'$ with $P' \mathcal{R} Q'$ (and symmetrically). Context bisimilarity $\sim$ is the largest context bisimulation; it contains $\equiv$.

Proposition 4 (Congruence, by construction). *If the GSLT minimal-context construction supplies relative pushouts for rho’s contexts [5, 6, 7], then context bisimilarity is a congruence: $P \sim Q$ implies $F[P] \sim F[Q]$ for every context F .*

Proof sketch. This is the Leifer–Milner theorem: when reactions are presented as idem-pushout labels over a category of contexts possessing relative pushouts, the derived bisimilarity is a congruence, because the bisimulation game already quantifies over the discriminating power of every context. The hypothesis — that rho’s contexts have the requisite relative pushouts — is the content of the GSLT minimal-context construction [7], and is recorded as Obligation 3. $\square$

This is the structural reason the GSLT programme labels by contexts: congruence is *built into* the equivalence rather than recovered afterwards. It settles, at the level of the definition, the point that an action-labelled presentation must negotiate by closing under name instantiation (compare Section 6).

Remark 5 (Commitments are the finite presentation). The offer-input clause of Definition 1 is a family $\{P \xrightarrow{[\cdot] \mid x!(Q)} P_0\{\text{@}Q/y\} \mid P_1 : Q \in \text{Proc}\}$ indexed by all messages. The family factors through a single *abstraction* $(y)(P_0 \mid P_1)$ (with $y \# P_1$): plugging the message $\text{@}Q$ into the abstraction gives the residual. Likewise the offer-output family factors through a *concretion*. These abstraction/concretion packagings, in Milner’s sense [4], are the finite presentation of the class-indexed context-transition families; we use them for the denotation of Section 5, with the understanding that the labels they abbreviate are the contexts of Definition 1.

3 Quoting is the colour-swap: a stratified reading

3.1 Four sorts and one involution

The construction of [2] takes two copies of one signature-generated set theory, red $\text{Set}[X]$ and black $\text{Set}[\bar{X}]$, and ties them by $X = \text{Set}[\bar{X}]$, $\bar{X} = \text{Set}[X]$: each colour’s atoms are the other’s sets. The

source paper reads these ties as identities, collapsing “red atom” with “black set”. For the rho model it is cleaner — and, as we will see, exactly the graded discipline [2], §6.5 calls for — to keep the sorts distinct and make the colour-moving maps first class. We name four sorts,

$$\mathbf{S} \text{ (red sets), } \mathbf{A} \text{ (red atoms), } \mathbf{S} \text{ (black sets), } \mathbf{A} \text{ (black atoms),}$$

and three families of maps.

The knots, as coercions. The two recursive knots become two isomorphisms,

$$\kappa: \mathbf{S} \xrightarrow{\cong} \mathbf{A} \quad (\text{black sets are red atoms; } X = \text{Set}[\bar{X}]),$$

$$\bar{\kappa}: \mathbf{S} \xrightarrow{\cong} \mathbf{A} \quad (\text{red sets are black atoms; } \bar{X} = \text{Set}[X]).$$

Where [2] writes $X = \text{Set}[\bar{X}]$ as an equality we write κ as a coercion: a black set, presented *as* a red atom, is red-opaque ([2], Remark 2); κ is the act of so presenting it.

The swap, an involution. The two copies are copies of one theory, so the construction is symmetric under exchanging the colours — the symmetry [2] invokes for its Corollary 2. We make it a map: a single polymorphic, *sort-preserving* involution

$$\mathbf{s}: \mathbf{S} \cong \mathbf{S}, \quad \mathbf{s}: \mathbf{A} \cong \mathbf{A}, \quad \mathbf{s} \circ \mathbf{s} = \text{id},$$

repainting red to black and back, structure for structure ($\mathbf{s}$ carries $\emptyset$ to $\emptyset$, $\cup$ to $\cup$, and so on). The swap commutes with the knots: the square

$$\mathbf{s} \circ \kappa = \bar{\kappa} \circ \mathbf{s} \quad (\text{both } \mathbf{S} \rightarrow \mathbf{A})$$

holds, expressing that repainting a black-set-as-red-atom agrees with repainting then re-presenting. This coherence is what lets one colour’s reflection generate the other’s.

3.2 The calculus’s reflection, derived

Take processes to be red sets and names to be red atoms. Define

$$\boxed{\ @ \ := \ \kappa \circ \mathbf{s} : \mathbf{S} \rightarrow \mathbf{A}, \quad * \ := \ \mathbf{s} \circ \kappa^{-1} : \mathbf{A} \rightarrow \mathbf{S}. }$$

Thus $@P$ first repaints the red process P black ($\mathbf{s}$), then presents the result as a red atom (κ): it “seals P into an atom of the opposite colour” exactly as [2], §9.3 proposes, the seal being the red-opacity of κ , the colour crossing being $\mathbf{s}$. The dereference $*x$ un-presents (κ^{-1}) and repaints back ($\mathbf{s}$).

Proposition 6 (Reflection is involutivity of the swap). *For every process P and name x ,*

$$*@P = \mathbf{s} \kappa^{-1} \kappa \mathbf{s}(P) = \mathbf{s} \mathbf{s}(P) = P, \quad @*x = \kappa \mathbf{s} \mathbf{s} \kappa^{-1}(x) = x.$$

Hence the model validates $@P \equiv P$ and $@*x \equiv_{\mathcal{N}} x$, not by imposing an equation but as the assertion that the colour-swap is an involution.*

Proposition 7 (Names implemented by swaps; the dual calculus). *Define the black calculus's reflection by $@_S := \bar{\kappa} \circ s : S \rightarrow A$ and $*_S := s \circ \bar{\kappa}^{-1}$. Then by the coherence $s\kappa = \bar{\kappa}s$,*

$$@_S = s \circ @ \circ s,$$

so the two colours' quotes are swap-conjugate: one reflection determines the other. The black calculus has black sets as processes and red sets as names; by [2], Corollary 2 the two interlock, each calculus's names being the other's processes. A single reflective universe carries two dual rho calculi at once.

Remark 8 (The stratification is the grading). [2], §6.5 calls for a graded modality separating colour-respecting (equivariant) computation from reflective computation, the price of breaking a name's opacity. The four sorts are that grading. A computation that stays in $\{S, A\}$ and uses only $@, *$ relabels red processes and names without ever observing black structure: this is the equivariant fragment, ordinary nominal/ π -style names with every guarantee intact. Applying s to compute with black set-operations on a name's interior is the reflective surplus, and the type — which sort one is in — records it. Decidable structural identity of names ([2], §6.5) is here the decidability of equality in A , settled out of red's sight by black equality of the swapped processes.

4 The behaviour functor and the process universe

The knotted universe is a model of nominal sets ([2], §6.1): the model M of its Theorem 1 carries the $\text{Sym}(\mathcal{A})$ -action with every element finitely supported and supports the name-abstraction $[\mathcal{N}]X = (\text{Name} \times X)/\sim$ with $\text{supp}((a)x) = \text{supp}(x) \setminus \{a\}$ ([2], §6.2). Write $\mathcal{U}$ for this nominal universe of red sets and identify $\text{Name} = A$ with $\text{Proc} = S$ along the reflection of Section 3.

Definition 9 (Behaviour functor). Let $B : \mathcal{U} \rightarrow \mathcal{U}$,

$$B(R) = \mathcal{P}(\partial T_K \times R),$$

the (powerset of) context-label/residual pairs of Definition 1, with ∂T_K the Fire context-labels (Remark 2). Because the input and output labels carry an unrestricted message or continuation, ∂T_K is class-indexed and B is taken with the full powerclass $\mathcal{P}$ rather than $\mathcal{P}_{\text{fin}}$; its final coalgebra accordingly lives in the class-sized framework of [2], §8.1, where it coincides with the colour-blind non-well-founded universe of [2], §7. The *commitment refinement* $B^b(R) = \mathcal{P}_{\text{fin}}(\text{Name} \times [\mathcal{N}]R \uplus \text{Name} \times \text{Name} \times R \uplus R)$ (input abstractions, output concretions, silent moves) presents the same bisimilarity finitely (Remark 5), assembled from constants, products, coproducts, finite powerset, and the abstraction functor $[\mathcal{N}](-)$, each preserving finite support and weak pullbacks; we use B^b when finite branching matters.

A B -coalgebra (R, β) equips every r with its set of context-labelled transitions — the data of Section 2.3. A coalgebra morphism is a finitely-supported map commuting with the structure maps: a functional context bisimulation.

Definition 10 (The process universe). $\text{Proc} := \nu B$, the final B -coalgebra in $\mathcal{U}$, with structure map $\langle \text{out}, \text{in}, \tau \rangle : \text{Proc} \rightarrow B(\text{Proc})$, and $\text{Name} := s(\text{Proc})$ presented as A via κ .

Unfolded, an element of νB over a state is the set of that state's transitions, the set of its residuals' transitions, and so on: the Aczel reading $\{(a, P') : P \xrightarrow{a} P'\}$ with Name -valued labels and an abstraction summand for input. A loop is non-well-founded descent under the colour-blind $\in_*$ of [2], §7.2 — a red process contains a black name whose red unfolding contains a red process, without end. The construction's global non-well-foundedness *is* the recursion of process behaviour, available for free where Aczel postulates it.

5 A Milner-style denotation

We now give $\llbracket - \rrbracket$ compositionally, following [4]. After Milner, a guarded process commits to an *action* carrying an *agent*: an input carries an *abstraction* $(y)P$ (it expects a name), an output a *concretion* $[Q]$ (it offers one). Communication is application.

Definition 11 (Commitment denotation). Define $\llbracket - \rrbracket$ on processes by

$$\begin{aligned} \llbracket \mathbf{0} \rrbracket &= \emptyset, & \llbracket \text{for}(y \leftarrow x) P \rrbracket &= \{ \langle \llbracket x \rrbracket^?, (y) \llbracket P \rrbracket \rangle \}, & \llbracket x!(Q) \rrbracket &= \{ \langle \llbracket x \rrbracket^!, [\llbracket Q \rrbracket] \rangle \}, \\ \llbracket *x \rrbracket &= * \llbracket x \rrbracket = \mathbf{s}(\kappa^{-1} \llbracket x \rrbracket), & \llbracket P \mid Q \rrbracket &= \llbracket P \rrbracket \parallel \llbracket Q \rrbracket, \end{aligned}$$

and on names $\llbracket @P \rrbracket = @ \llbracket P \rrbracket = \kappa(\mathbf{s} \llbracket P \rrbracket) \in \mathbf{A}$. The tags $\llbracket x \rrbracket^?$ and $\llbracket x \rrbracket^!$ abbreviate the context labels $[\cdot] \mid x!((-)$ and $[\cdot] \mid \text{for}((-) \leftarrow x)(-)$ of Definition 1, in their finite commitment presentation (Remark 5): the abstraction $(y) \llbracket P \rrbracket \in [\mathcal{N}] \nu \mathbf{B}$ binds the received name, packaging the class-indexed family of input context-transitions, and the concretion $[\llbracket Q \rrbracket]$ carries the message $\llbracket Q \rrbracket \in \nu \mathbf{B}$. Each guarded prefix denotes a *singleton* commitment, one cut at which P meets its environment.

Definition 12 (Reaction; the swap fires). On commitments, application is

$$(y) R \cdot [M] = R\{ @M / y \} = R\{ \kappa(\mathbf{s} M) / y \},$$

and reaction between dual commitments on equivalent names is

$$\frac{\langle a^?, (y) R \rangle \in \llbracket P \rrbracket \quad \langle a^!, [M] \rangle \in \llbracket Q \rrbracket \quad a \equiv_{\mathcal{N}} a'}{\llbracket P \rrbracket \mid \llbracket Q \rrbracket \rightsquigarrow R\{ @M / y \} \parallel (\text{residuals})} \text{REACT}.$$

The message M , a red set (a process), is sealed by $@ = \kappa \circ \mathbf{s}$ into the name $@M$ bound by the receiver: *communication is precisely where the colour-swap of Section 3 fires*. This is the set-theoretic content of rho's COMM: the receiver learns not the process sent but its name.

Definition 13 (Semantic parallel). $\parallel : \nu \mathbf{B} \times \nu \mathbf{B} \rightarrow \nu \mathbf{B}$ is defined corecursively (by finality) as the expansion

$$\begin{aligned} R \parallel R' &= \{ \langle \alpha, T \parallel R' \rangle : \langle \alpha, T \rangle \in R \} \cup \{ \langle \alpha, R \parallel T' \rangle : \langle \alpha, T' \rangle \in R' \} \\ &\cup \{ \langle \tau, U \parallel U' \rangle : \langle a^?, (y) U \rangle \in R, \langle a^!, [M] \rangle \in R', a \equiv_{\mathcal{N}} a', U = (\text{redex residual}) \} \end{aligned}$$

together with the symmetric τ -set (a $?$ -commitment of R' with a $!$ -concretion of R). Here the silent reactions apply REACT (Definition 12), so U is $U_0\{ @M / y \}$ for the abstraction body U_0 . On immediate commitments $\parallel$ restricts to union, recovering the Milner intuition $\llbracket P \mid Q \rrbracket \supseteq \llbracket P \rrbracket \cup \llbracket Q \rrbracket$; but the residuals carry the surviving context $T \parallel R'$, so $\parallel$ is not bare union. This is what keeps $a!(0) \mid a!(0) \not\approx a!(0)$: the first, after one $a!()$, retains the second output, where bare union would discard it.

Proposition 14 (The two denotations agree). *The compositional $\llbracket - \rrbracket$ of Definitions 11–13 coincides with the unique $\mathbf{B}^b$ -coalgebra morphism from the syntactic coalgebra $(\mathcal{T}/\equiv, \partial)$ of Section 2.3 into $\nu \mathbf{B}^b$, where ∂ records each state's context-transitions in their commitment presentation (Remark 5).*

Proof sketch. Both are $\mathbf{B}^b$ -coalgebra morphisms out of $(\mathcal{T}/\equiv, \partial)$: the clauses of Definition 11 read off exactly the commitments ∂ assigns (singleton for each prefix, $\emptyset$ for $\mathbf{0}$), and $\parallel$ is the corecursive map realising PAR and COMM/REACT. By finality the morphism is unique, so the two agree. $\square$

Theorem 15 (Adequacy and the reflection law). *For all processes P, Q and names x :*

- (i) $\llbracket @P \rrbracket = @(\llbracket P \rrbracket)$, $\llbracket *x \rrbracket = *x$, and $\llbracket *@P \rrbracket = \llbracket P \rrbracket$: *the reflection law holds in the model, as $\mathbf{s} \circ \mathbf{s} = \text{id}$ (Proposition 6);*
- (ii) $\llbracket - \rrbracket$ *is a homomorphism: $\llbracket \mathbf{0} \rrbracket = \emptyset$, $\llbracket P \mid Q \rrbracket = \llbracket P \rrbracket \parallel \llbracket Q \rrbracket$, and the prefix clauses of Definition 11; in particular $P \equiv Q \Rightarrow \llbracket P \rrbracket = \llbracket Q \rrbracket$, the commutative-monoid laws and reflection being special cases.*

6 Full abstraction

We prove that $\llbracket - \rrbracket$ identifies exactly the context-bisimilar processes. The argument is the standard coalgebraic one over the commitment refinement $\mathbf{B}^b$, which finitely presents context bisimilarity; we then read the result back through OSLF adequacy and note that, unlike the action-labelled case, congruence is already in hand.

Lemma 16 (Context = coalgebraic bisimilarity). *Context bisimilarity (Definition 3) coincides with $\mathbf{B}^b$ -coalgebraic bisimilarity. For the relational lifting $\overline{\mathbf{B}}^b$, a relation $\mathcal{R}$ on $\mathcal{T}/\equiv$ is a coalgebra bisimulation iff it is a context bisimulation.*

Proof sketch. By Remark 5 the commitment presentation packages each class-indexed context-transition family as a single abstraction or concretion, with no loss of discriminating information: two states pass the same context labels iff their commitment sets match. $\overline{\mathbf{B}}^b$ relates behaviours when their output and silent summands match with $\mathcal{R}$ -related residuals and their input summands match in $[\mathcal{N}]\mathcal{R}$; unfolding $[\mathcal{N}]\mathcal{R}$ by its defining equivalence $((y) R [\mathcal{N}]\mathcal{R} (y) R' \text{ iff } R\{a/y\} \mathcal{R} R'\{a/y\})$ for all names a) reproduces the matching of the offer-input context family, and the concretion case the offer-output family. Hence the two notions of bisimulation coincide. $\square$

Lemma 17 ($\mathbf{B}^b$ is well behaved). *$\mathbf{B}^b$ preserves weak pullbacks. Hence on any $\mathbf{B}^b$ -coalgebra (C, γ) , coalgebraic bisimilarity is the kernel of the unique morphism $!_C: C \rightarrow \nu \mathbf{B}^b$: $c \sim_{\mathbf{B}^b} d \iff !_C(c) = !_C(d)$.*

Proof sketch. $\mathcal{P}_{\text{fin}}$, products, coproducts and constants preserve weak pullbacks, and so does the nominal abstraction functor $[\mathcal{N}](-)$ (its quotient by the freshness equivalence is by a functional, hence pullback-stable, relation); weak-pullback preservation is closed under these constructions, so $\mathbf{B}^b$ preserves them. For a weak-pullback-preserving functor with a final coalgebra, behavioural equivalence (the kernel of $!_C$) coincides with coalgebraic bisimilarity — the standard fact of universal coalgebra [10]. Existence of $\nu \mathbf{B}^b$, and its agreement with the class-sized $\nu \mathbf{B}$ on equality, is Obligation 1. $\square$

Theorem 18 (Full abstraction for context bisimilarity). *For all rho processes P, Q ,*

$$\llbracket P \rrbracket = \llbracket Q \rrbracket \iff P \sim Q.$$

Moreover, since $\sim$ is a congruence (Proposition 4), this is full abstraction with respect to a congruence: $\llbracket - \rrbracket$ identifies P and Q exactly when no context can tell them apart.

Proof. By Proposition 14, $\llbracket - \rrbracket$ is the unique morphism $!: (\mathcal{T}/\equiv, \partial) \rightarrow \nu \mathbf{B}^b$, so $\llbracket P \rrbracket = \llbracket Q \rrbracket$ iff $!(P) = !(Q)$. By Lemma 17 on the syntactic coalgebra, $!(P) = !(Q)$ iff P, Q are coalgebraically bisimilar; by Lemma 16 that is context bisimilarity $\sim$. Chaining, $\llbracket P \rrbracket = \llbracket Q \rrbracket \iff P \sim Q$. Congruence is Proposition 4. $\square$

Corollary 19 (Adequacy through OSLF). *Let $\mathcal{L}$ be the Hennessy–Milner logic the OSLF construction generates from the context-labelled transition system, with modalities $\langle F \rangle \varphi$ indexed by the labels ∂T_K (Remark 2). By OSLF adequacy [7], $P \sim Q$ iff P and Q satisfy the same formulae of $\mathcal{L}$. Combining with Theorem 18,*

$$\llbracket P \rrbracket = \llbracket Q \rrbracket \iff P \sim Q \iff \forall \varphi \in \mathcal{L}. (P \models \varphi \Leftrightarrow Q \models \varphi).$$

Equality in the model, context bisimilarity, and modal indistinguishability are one relation seen three ways — the Firing, Placing, and Witnessing readings of the single label set ∂T_K .

Corollary 20 (Internal full abstraction; Aczel as a fragment). *On $\nu \mathbf{B}^b = \mathbf{Proc}$ itself, two elements are context-bisimilar iff equal: behavioural equivalence is the equality of the model. Aczel’s strong extensionality for CCS is recovered as the special case where the labels ∂T_K degenerate to an opaque set of action names over which no further context structure is observed — the equivariant fragment of Section 7.*

What the context labels bought. An action-labelled presentation would here have to negotiate congruence as an afterthought: strong bisimilarity in name-passing calculi may fail to be preserved by input prefix unless one closes under name instantiation, replacing $\sim$ by the largest bisimulation-congruence $\sim^c$. Labelling by contexts removes the negotiation. The bisimulation game of Definition 3 already ranges over every environment, so Proposition 4 delivers congruence directly (granted the relative pushouts of Obligation 3), and Theorem 18 is full abstraction for a congruence with no closure step. Rho’s structure makes the relative-pushout hypothesis especially plausible: there is no ν , so no scope-extrusion interaction with the cut, and a name is structured data, so name-for-name substitution is set-theoretic substitution under $\kappa \circ \mathbf{s}$ with no fresh-name side conditions.

7 The risk ledger

The accounting of [2] transfers, and locates where rho’s recursion is paid for.

Finite processes: the hereditarily-finite core, no infinite risk. A process with a finite synchronisation tree is a hereditarily-finite red set with finitely many black-set atoms (its free names), living in the finitary core of [2], §5.5, where the final coalgebra of the finite-powerset functor is literal and exists at set size. Foundation and finite support are theorems, not axioms; the commitment is the black empty set and one successor, no more.

Recursion: one ω , black infinity, no primitive names. Recursion enters rho through reflection (Section 2): a finite term can loop by quoting and dereferencing its own continuation. Such a process is a non-well-founded inhabitant of $\nu \mathbf{B}$, available only in the infinitary reading, which by [2], §5.6 and §8.1 costs exactly one ω — black infinity — and lives, for the full powerset, in the class-sized final-coalgebra framework. As with the completed datatype Λ of [2], §6.2, each finite-state recursive process is set-sized while the universe of all processes is the one- ω object. No primitive name is ever reintroduced; names remain black sets built from $\emptyset$.

Aczel’s CCS model as the equivariant shadow. Stay within one colour, never apply $\mathbf{s}$ to a name’s interior, treat names as opaque swap-only atoms (the equivariant fragment, [2], §6.5). The input summand’s abstraction degenerates, $\mathbf{B}$ reduces to $\mathbf{B}_0(R) = \mathcal{P}_{\text{fin}}(\text{Act} \times R)$, and $\nu \mathbf{B}_0$ is Aczel’s CCS model (Corollary 20). Aczel’s construction is the colour-respecting fragment; the rho surplus

— structured, computed, content-addressed names, and the licence to cross the colour barrier with
 $*$ — is the reflective remainder, graded by the four sorts of Section 3.

8 Toward a rigorous construction

In the manner of [2], §8, the obligations a fully rigorous development would discharge. For the finitary core each reduces to a statement about hereditarily-finite sets, and several are corollaries of the model built.

Obligation 1 (Existence of the final coalgebra, and $\nu\mathbf{B}^b \sim \nu\mathbf{B}$). Exhibit the final coalgebras via the two-sorted endofunctor $H(R, B) = (B + \mathcal{P}_s R, R + \mathcal{P}_s B)$ of [2], §8.1, taken at the *final* coalgebra (greatest fixed point). For the commitment refinement $\mathbf{B}^b$, finite branching makes it accessible and $\nu\mathbf{B}^b$ exists in the nominal universe, at set size on the finitary core (the limit of the terminal ω^{op} -chain). For the full context functor $\mathbf{B} = \mathcal{P}(\partial T_K \times -)$, the final coalgebra is the class-sized, colour-blind non-well-founded universe of [2], §7. Show the two carry the same equality — that the commitment packaging of Remark 5 is a behavioural equivalence — so bisimilarity computed in $\nu\mathbf{B}^b$ is context bisimilarity in $\nu\mathbf{B}$.

Obligation 2 (The stratified signature and its coherence). Fix the four-sorted signature of Section 3 — the sorts **S**, **A**, **S**, **A**, the knots $\kappa, \bar{\kappa}$, the swap **s** — as an extension of the $\text{Set}[X]$ signature of [2], and verify the equational theory: **s** an involution and a homomorphism for all set operations, $\mathbf{s}\kappa = \bar{\kappa}\mathbf{s}$, and $\kappa, \bar{\kappa}$ isomorphisms. Then $@, *$ are derived and Proposition 6 is a calculation. The collapse of [2] — $\kappa, \bar{\kappa}$ literal identities — is recovered as a quotient of this theory, and one checks the reflection law survives the quotient.

Obligation 3 (Relative pushouts for rho’s contexts). Discharge the hypothesis of Proposition 4: that the category of rho contexts, with plugging as composition, possesses relative pushouts, so the idem-pushout labels of Definition 1 are well defined and the Leifer–Milner congruence theorem applies. This is the GSLT minimal-context construction [7]; the cut lying always at a parallel composition (no reaction under a prefix), and names being quoted structure rather than ν -bound, are the features that should make the relative pushouts exist and be computable. With this discharged, $\sim$ is a congruence outright, and the action-labelled closure-under-substitution question does not arise.

Obligation 4 (Context correspondence and OSLF adequacy). Promote Lemma 16 to a theorem: define the relational lifting $\bar{\mathbf{B}}^b$ precisely, including the abstraction summand, and prove context bisimilarity equals $\bar{\mathbf{B}}^b$ -bisimilarity. Relate $\sim$ to reduction-barbed congruence so that Theorem 18 is full abstraction for the intended equivalence; rho’s lack of ν removes the closure under restriction this normally requires. Finally, instantiate the OSLF construction on the context-labelled system and prove the adequacy used in Corollary 19, so that the modalities $\langle F \rangle \varphi$ indexed by ∂T_K characterise $\sim$.

Obligation 5 (Substitution and the swap-firing reaction). Prove $\langle P\{ @M/y \} \rangle = \langle P \rangle \{ @ \langle M \rangle / y \}$, i.e. that $\langle - \rangle$ is a map of nominal algebras for substitution and that the application of Definition 12 computes it, with the seal $\kappa \circ \mathbf{s}$ applied to the message. This is the lemma underwriting REACT and clause (ii) of Theorem 15 for input, carried by the abstraction functor’s evaluation $[\mathcal{N}]R \times \text{Name} \rightarrow R$.

Obligation 6 (Barbed-congruence calibration). With congruence secured by Obligation 3, calibrate which behavioural equivalence the model realises: confirm that context bisimilarity $\sim$ coincides with strong barbed congruence for rho, so that Theorem 18 is full abstraction for the equivalence

practitioners use. Rho’s structured, ν -free names are again the reason to expect the calibration to be tight where in π it is delicate.

Obligation 7 (Placing the construction). Locate the model among Aczel’s CCS model [1] (the equivariant fragment), the presheaf and nominal coalgebraic models of name-passing (Fiore–Turi [13], Stark [14]), of which this is the reflective, ν -free instance with $\mathbf{Name} \cong \mathbf{Proc}$, and the bialgebraic semantics of Turi–Plotkin [15], into whose abstract-GSOS format the rho rules fit. The defining feature absent from all three — $\mathbf{Name} \cong \mathbf{Proc}$ via $\mathbf{s}$ — is the second knot.

9 Conclusion

Rho’s defining move — names are quoted processes, $*@P \equiv P$ — is the colour-swap of the knotted universe, read through a stratification into four sorts: $@ = \kappa \circ \mathbf{s}$, $* = \mathbf{s} \circ \kappa^{-1}$, and the reflection law is the involutivity $\mathbf{s} \circ \mathbf{s} = \text{id}$, true by construction. The dynamics follow Milner: a process is its set of commitments, input an abstraction and output a concretion, communication their application — and the application is where the swap fires, sealing the sent process into the received name. The denotation lands in the final coalgebra of a behaviour functor over the knotted nominal universe, where the colour-blind non-well-foundedness supplies recursion for free and finality makes bisimilar processes equal. Following the GSLT discipline we labelled transitions by contexts — the minimal environments in which a process completes a reaction, which are at once the program/environment cuts, the OSLF modality indices, and the locations in term structure — and expressed bisimulation over them; on that base we proved full abstraction, $\langle\!\langle P \rangle\!\rangle = \langle\!\langle Q \rangle\!\rangle \iff P \sim Q$. The context labelling pays a structural dividend the action-labelled account could not: the equivalence is a congruence by construction, so full abstraction holds for a congruence with no closure step, granted only that rho’s contexts admit relative pushouts — a hypothesis its ν -free, quoted-name structure makes especially plausible. The risk ledger is the source paper’s: finite processes cost nothing beyond the empty set, recursion costs one ω , and not one primitive name is ever spent. Aczel’s model sits inside this one as its equivariant shadow; the surplus is reflection, and reflection is the swap.

Acknowledgements

This note develops Section 9.3 of *A Knotted Universe*, with the stratified reading of the colour barrier, the Milner-style commitment semantics over context-labelled transitions, and the full-abstraction theorem as its additions.

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